MAJOR PROJECT · INDIAN CRIMINAL LAW · LEGAL AI

Pramana

प्रमाण *proof; a means of valid knowledge*

An AI legal research assistant for Indian criminal law that checks every section it cites, across the switch from the IPC to the BNS. It runs entirely on a laptop.

✓ Every citation verified | Fully offline | Zero cost

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OVERVIEW

What we will cover.

01

The problem

A renumbered criminal code, and AI that invents law

02

Our solution

What Pramana does, from facts to a checked report

03

How it works

The pipeline, the citation graph and the bridge from old law to new

04

Technology

The stack, and how it fits on a 4 GB laptop GPU

05

Live demo

A real case, run end to end in the browser

06

Results and next steps

What we measured, and where it goes from here

On 1 July 2024 India renumbered its criminal law.

INDIAN PENAL CODE, 1860 offences before 1 July 2024	SECTION	BHARATIYA NYAYA SANHITA, 2023 offences from 1 July 2024
Punishment for murder	302	Uttering words with deliberate intent to wound religious feelings
Punishment for murder by a life-convict	303	Theft
Cheating and dishonestly inducing delivery of property	420	*No such section. The BNS ends at 358.*

A section number means nothing without its code, and *the offence date decides the code.*

AI models cite law with confidence, including law that does not exist.

A 4-BILLION-PARAMETER MODEL, ASKED ON THIS LAPTOP

Q What does BNS Section 303 cover?

A A public-panic offence: making or spreading a bomb threat.

× Wrong. BNS 303 is Theft.

qwen3.5:4b, the model inside Pramana. Answer paraphrased.

20 *of* **68**

citations named sections that do not exist in the BNS when the model answered from memory, mostly old IPC numbers.

AND THE FACTS ARE CONFIDENTIAL

A lawyer types a client's real facts. They should never be sent to a cloud AI service, so the whole system has to run on the lawyer's own machine.

Facts and a date go in. A checked research memo comes out.

CASE FACTS

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 Aug 2024

about a minute, offline

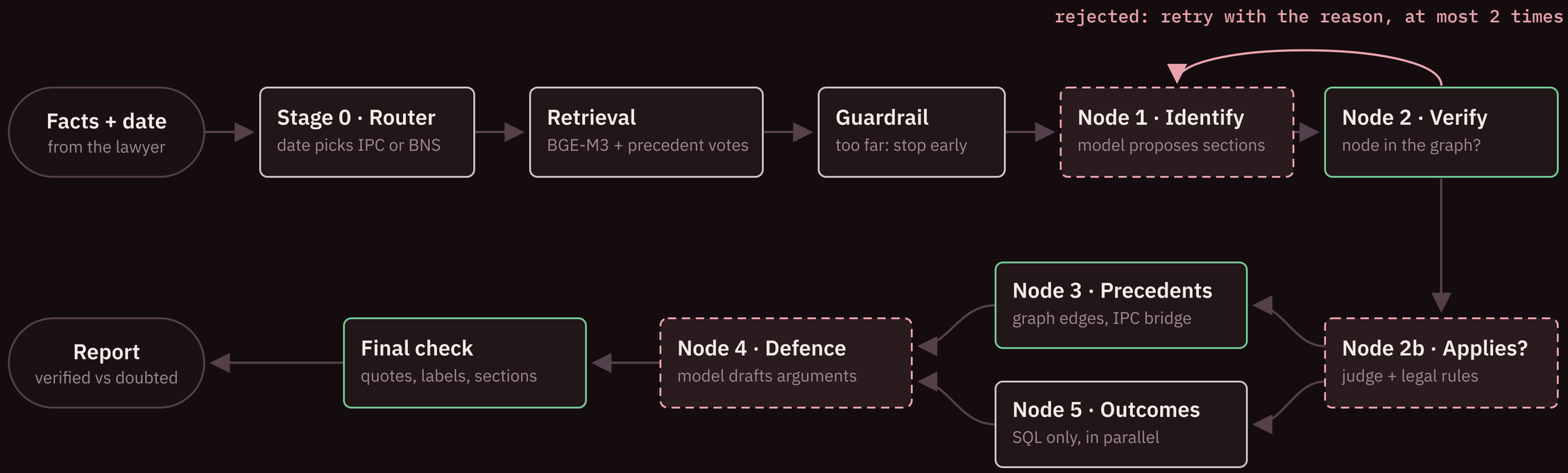
CODE	BNS, because the offence is on or after 1 July 2024
CHARGE	BNS 303 Theft ✓ verified
PRECEDENTS	3 judgments, found through its IPC predecessor, IPC 379
DEFENCE	A counter-argument, its quote checked against the facts
OUTCOMES	How 158 appeals under IPC 379 were decided

Verified citations
a section is shown only if it exists in the right code

Private by design
runs offline; client facts never leave the laptop

Free to run
open models and public data; no API key or account

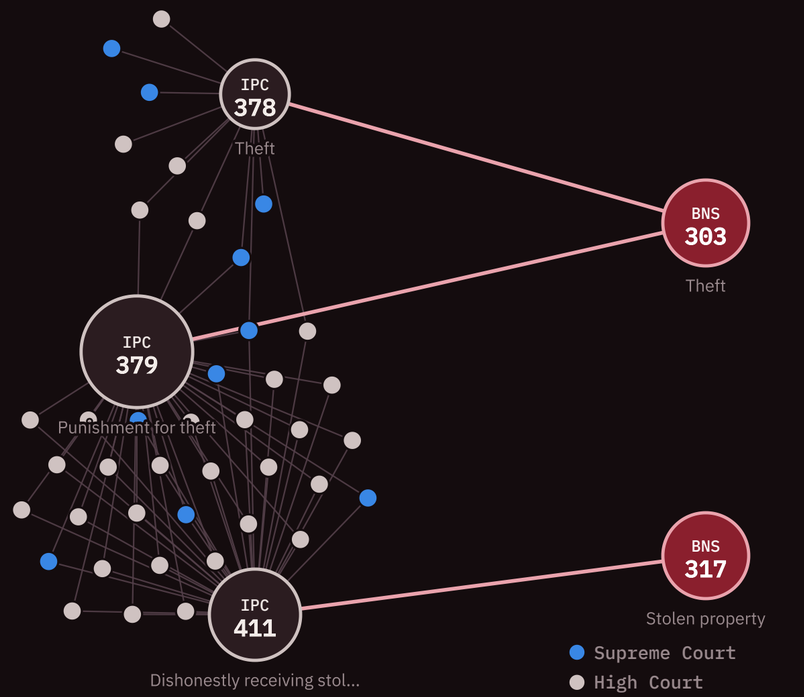
The model proposes. The graph decides.



Deterministic code · Language model · Graph lookup

BUILT OFFLINE · ChromaDB · 4 collections · Citation graph · NetworkX · Outcomes · SQLite

Every citation is checked against the law itself.



933 sections · 2,699 judgments · 16,493 citation edges · 539 IPC→BNS bridges

THE RETRY LOOP · A REAL RUN WITH RETRIEVAL WITHHELD

ROUND 0 — model: 421

✕ Not a section of the BNS, which has sections 1 to 358.

ROUND 1 — model: 378

✕ An IPC number. Hint from the graph: its BNS counterpart is **BNS 303**.

ROUND 2 — model: 303

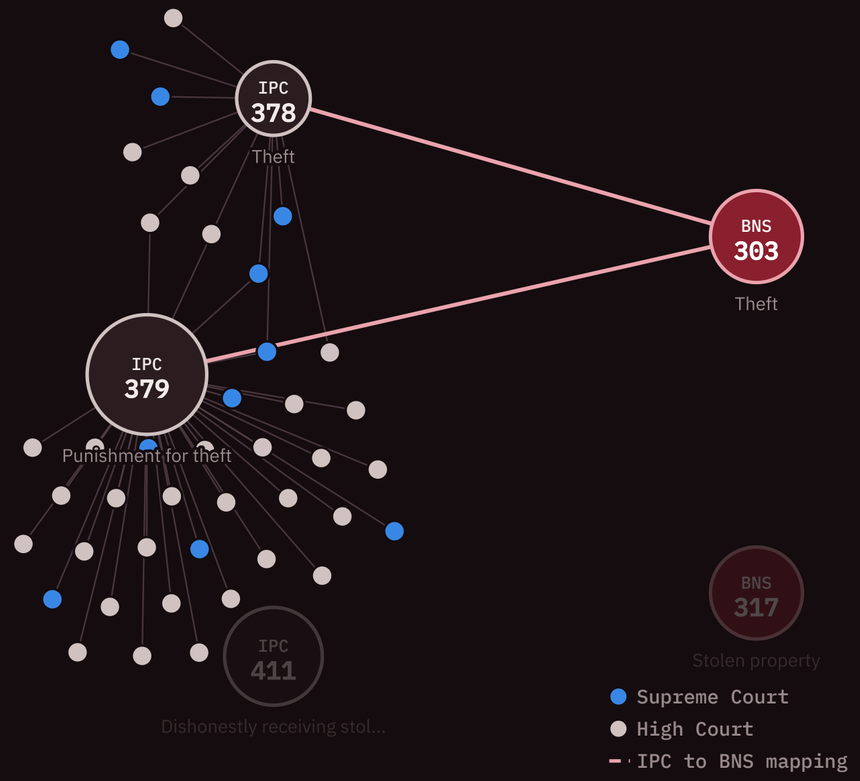
✓ **BNS 303 · Theft.** A node of the BNS. Verified.

20 / 20
real vs invented section numbers told apart

0
rejected citations ever reached a report

> Then a second check: does each real section fit the facts? Doubts are shown to the lawyer, never hidden.

New law has almost no case law yet. We bridge it to the old.



BNS 303	0	judgments cite it directly
IPC 378	11	via the mapping
IPC 379	158	via the mapping

TOP 3 BY SIMILARITY TO THE FACTS

P1	Telangana High Court	0.686	Allowed
P2	Himachal Pradesh High Court	0.676	Allowed
P3	Delhi High Court	0.658	Dismissed

Open-source, end to end, on one laptop.

INTERFACE	**FastAPI** backend · **Web app** live pipeline view · Streamlit
ORCHESTRATION	**LangGraph** state machine · retry edge · parallel nodes
INTELLIGENCE	**qwen3.5:4b** via Ollama, local · **BGE-M3** embeddings, GPU
KNOWLEDGE	**ChromaDB** 87,036 passages · **NetworkX** citation graph · **SQLite** outcomes
DATA	**IPC** 575 sections · **BNS** 358, official gazette · **2,745** SC + HC judgments · IPC→BNS map

ONE LAPTOP · RTX 3050 TI · 4 GB GPU



BGE-M3 encodes the facts · then the language model

Both models cannot fit at once, so every run hands the card over: unload the language model, embed, free the embedder, generate.

₹0
recurring cost

422
automated tests

BGE-M3 was chosen by a measured bake-off: recall@3 0.533 against 0.20 for a legal-domain model.

05 · LIVE DEMO

Let's run it.

THE CASE

The complainant parked his motorcycle outside his house at night. In the morning it was gone. CCTV shows the accused pushing it away down the lane and riding off. The accused had no permission to take it and did not return it.

OFFENCE DATE 15 August 2024

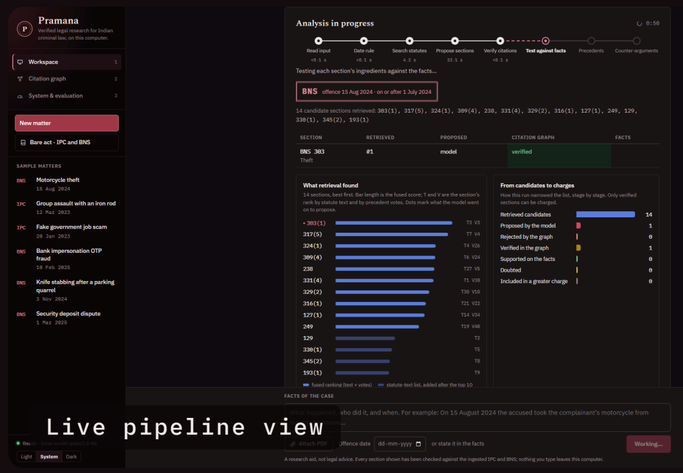
WATCH FOR router picks BNS · 303 ranked first · verified in the graph · precedents via IPC 379

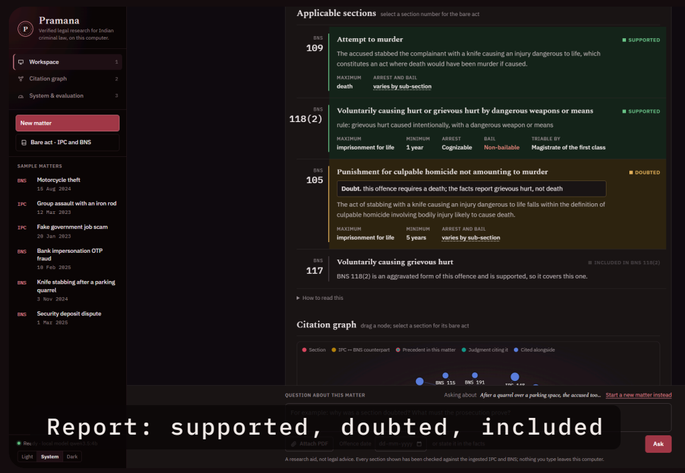
▶ LIVE

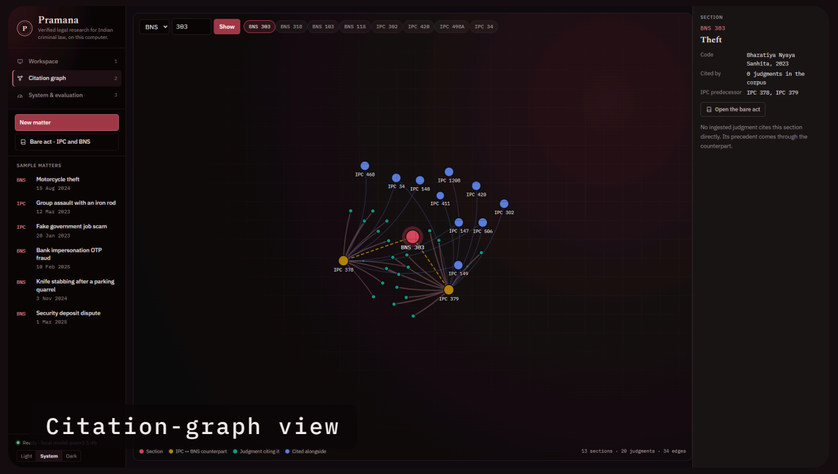
Switching to the browser

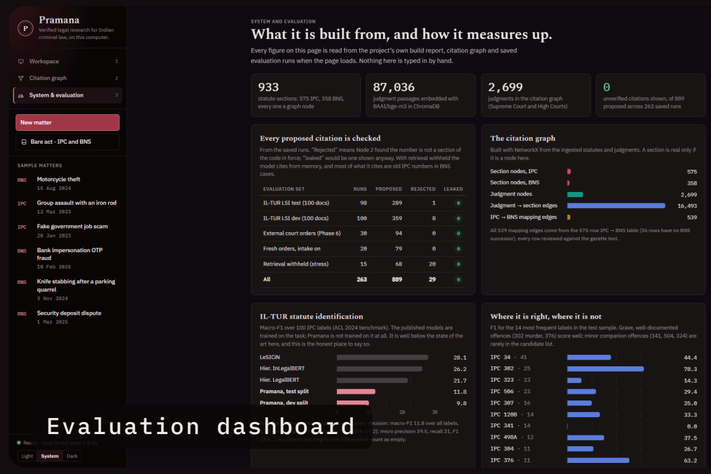
http://127.0.0.1:8000

Verified and doubted are never confused.









0

invented or wrong-code sections reached a report, across every evaluation we ran.

20 / 20
citation gate: real vs invented section numbers

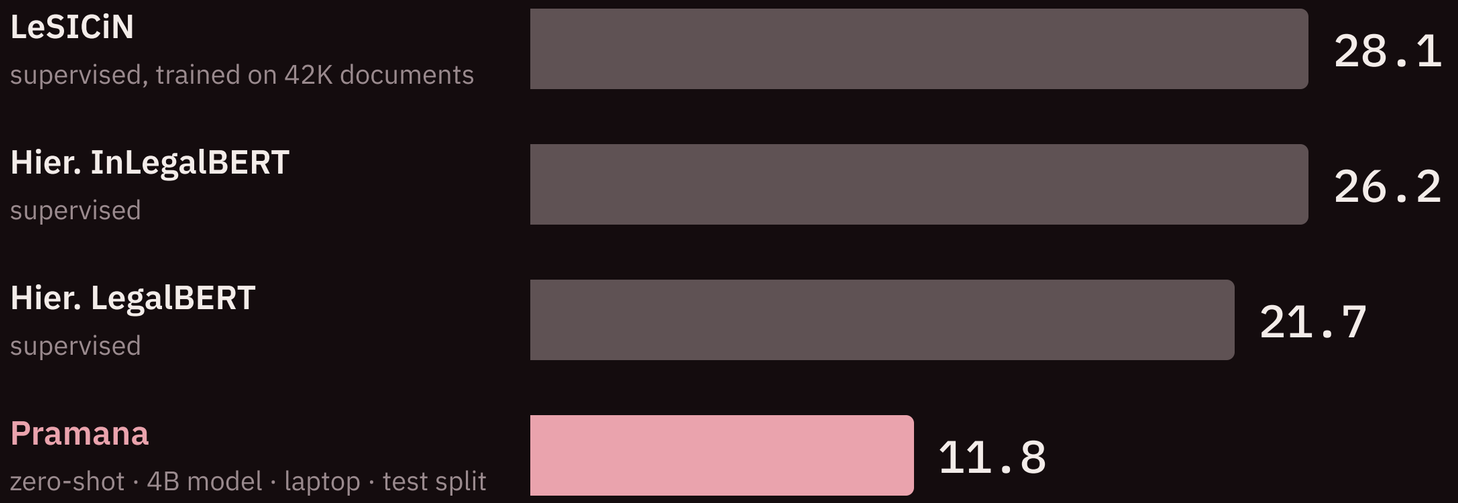
30 / 30
routing: the right code every time

889
citations checked across 263 saved runs; none unverified shown

6 / 6
cases recovered within 2 retries when the model cited from memory

Benchmarked on India's published standard, with no task-specific training.

LEGAL STATUTE IDENTIFICATION · MACRO-F1 OVER 100 LABELS



STRONG ON SERIOUS OFFENCES

302	Murder	0.78
376	Rape	0.63
34	Common intention	0.44

THE NEXT GAIN: WIDER RETRIEVAL

Minor companion offences such as simple hurt (323) or insult (504) rarely reach the candidate list: 37% of the right sections did. Widening retrieval is the clearest next step.

100 test documents, run once after all development. Development split: 9.8. Micro precision / recall / F1: 40 / 21 / 27.

Built for the transition years, and ready to grow.

WHO IT HELPS

Advocates
first-pass research across the IPC and the BNS

Students and legal-aid clinics
free, with the bare act and an IPC-to-BNS comparison

Firms with confidential matters
nothing leaves the office machine

Any field where citations must be real
the verify-against-a-graph pattern carries over

ROADMAP

DONE
Working system: seven phases, web app, evaluation

NEXT
Wider retrieval for minor companion offences

NEXT
Review of the IPC→BNS mapping by practising lawyers

NEXT
New BNS judgments ingested as courts publish them

LATER
BNSS and BSA, Hindi input, scanned orders

Thank you.

Pramana: every section it shows exists in the code the offence date selects, and it says so when it doubts one.

RAG, not fine-tuning | BGE-M3 · ChromaDB · NetworkX · LangGraph | qwen3.5:4b on Ollama | FastAPI

Suvan Sharma · The Spartan Coders · IKG PTU · Mentor: Aditya Sharma

Questions welcome.